Industrial Technology 50700

Measurement and Evaluation in Industry and Technology

# Greenhouse Gas Emissions: Passenger Cars Vs. SUVs

## Executive Summary

Since the last quarter of the twentieth century, the United States has sought to reduce oil consumption and carbon dioxide (CO2) emissions from automobiles. CO2 is a greenhouse gas (GHG) that has been proven to contribute to warming the global temperature to unstable levels resulting in climate change. The transportation industry contributes the most GHG emissions globally, making this sector a target for reduction measures. Through regulatory policies and innovative solutions in the auto industry, CO2 emissions from automobiles have decreased and fuel efficiency has increased. Over the years vehicles in almost every category have seen improvements in their emissions through innovations in the industry. However, the auto market has shifted away from passenger cars to larger SUVs and trucks. Despite the gains in CO2 emission reduction over the last several decades, the shift to larger commercial vehicles has and will continue to offset these gains unless effectively mitigated.

This paper compares CO2 emissions from gasoline powered passenger vehicles to that of gasoline powered sports utility vehicles (SUVs) that have gained popularity over the traditional sedan model car. A sample of sedan cars, SUVs, their respective CO2 emissions and vehicle weights will be analyzed and compared. The data and information presented in this paper will be used to test the hypothesis that having a higher proportion of SUVs in use will result in higher levels of CO2 emissions. This of course will continue to offset any gains made in CO2 reduction from automobiles overall.

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## Background

Greenhouse gases (GHG) in the atmosphere have become a major concern over the last few decades. These gases trap heat in the earth's atmosphere and are a major contributor of global climate change. Carbon dioxide (CO2) is produced when fossil fuel is burned during the operation of internal combustion engines. CO2 is one of the GHGs that has contributed to some of the most extreme climate disturbances. As the population becomes more concerned with the environment, governments and regulatory agencies have been charged with developing preventative measures. One such environmental goal is reaching net zero emissions by 2050, initiated by the Intergovernmental Panel on Climate Change (IPCC). This measure, along with other regulations that predate it have aided and influenced the ever-evolving fuel efficiency and emissions over the years.

Since the mid-seventies, enhancing fuel efficiency and lowering carbon output have been the focus of the automotive industry. The US government has focused primarily on fuel economy to help reduce CO2 output from automobiles. The corporate average fuel economy standards (CAFÉ) were established in 1975 [1]. Through these standards and innovative measures in the automotive industry CO2 output from automobiles has steadily decreased. “Since 2004, CO2 emissions have decreased by 27% and fuel efficiency has seen a 35% increase” [3]. However, despite the gains in fuel economy over the decades, the automotive industry’s marketing of less efficient vehicles threatens to offset any achievements in emissions reduction.

“In 2022, 37% of all new vehicles were sedan cars, while 63% of new vehicles were SUVs/trucks” [3]. Sedans can be defined as cars that typically have four doors, with the engine, cargo and passengers in separate compartments. SUVs are larger and weigh more than a standard car, resulting in these vehicles producing more CO2 emissions and having lower fuel efficiency.

The SUV can be defined as four-wheel drive vehicles that have the most capabilities of a light truck such as off roading and towing [4]. One of the issues with SUVs is that they stay in service twice as long as standard cars. According to research, "32.4% of SUVs will still be in use after 20 years compared to 15.7% of cars" [1]. The presence of older model SUVs on the road will compound the emission of CO2 seeing as the older vehicles are less fuel efficient than recent models. "Over a forty-two-year period (1975-2017) passenger car shares have shrunk from over 80% to 41%, while light trucks/SUVs increased from 19.4% to 59%" [1]. This increase in SUVs reduces the overall fuel economy of light duty vehicles and increases CO2 emissions.

The main factor in the automotive industry shift towards SUVs is marketing. The intentional decisions by automakers to heavily advertise SUVs and trucks has resulted in their overwhelming dominance in the auto market. An example of this: Chevrolet spent over $100 million promoting the 2017 Silverado compared to the $15 million advertising for the Chevy Bolt [1]. Many sedan models have been discontinued due to this marketing tactic. General Motors has ceased production of the Chevrolet Impala, Cruise, Cobalt, and Sonic in recent years, leaving buyers with the Chevrolet Malibu as the sole sedan option. Other automakers offer similar options to buyers, leaving them a variety of larger vehicles with higher CO2 output and lower fuel economy.

The market shift toward larger vehicles has the potential to offset a majority of the gains the auto industry has achieved in CO2 emissions reduction and efficiency. GHGs continue to increase annually, and the transportation sector has been the largest source of GHGs in the United States. With the growing proportion of SUVs in operation, this trends will continue unless effectively mitigated through innovative technology and techniques.

## Methods

The data on CO2 emissions was ascertained through fuel economy.gov and is given in grams per mile. CO2 emissions (g/mile) were recorded for standard 4-cylinder engine passenger cars (sedans), 4-cylinder engine SUVs, and 6-cylinder SUVs. Ten samples of each category were taken from five model years 2020-2024 for comparative analysis: fifteen sets of data total. The CO2 output from sedans with 4-cylinder engines is compared to CO2 output from 4-cylinder engine SUVs and then to 6-cylinder engine SUVs. The weight of each type of vehicle was also recorded and compared across all three categories. This information was ascertained from car and driver.com.

Statistical analysis of the mean, standard deviation and variance of CO2 emissions will be conducted. The mean(average) CO2 emissions of 4-cylinder cars for each year will be compared to the mean CO2 emissions from 4-cylinder SUVs and then to 6-cylinder SUVs for each corresponding year. The variance and standard deviation of each data set will be evaluated using the formulas:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} \qquad s = \sqrt{s^2}$$

Variance Standard Variation

Major factors affecting carbon emissions will be analyzed such as vehicle weight, and engine size. The mean average of a standard passenger car vehicle weight will be compared to that of SUVs. A hypothesis test will also be conducted to test the hypothesis that SUVs will produce more carbon emissions and thus continue to offset any and all gains made in GHG mitigation. An ANOVA test will be used to analyze the sample mean CO2 emission of each category.

## Data Analysis

Table 1 displays average CO2 emissions and weight for passenger car model years 2020-2024. The mean CO2 output (measured in grams per mile) from sedan cars is considerably lower than CO2 emissions from SUVs. An improvement in emissions can be observed as the model years progress from 2020 to 2024. Vehicle weight over the five-year span remains stable, with minimal increases. Overall, the average vehicle weight for standard sedan passenger cars is considerably lower than that of SUVs (when compared to tables 2 & 3). This is consistent with data that suggests that heavier vehicles will produce more carbon emissions.

**Table 1: CO2 Emission Data of 4-Cylinder Sedan Models**

| Year | Avg. CO2 Emissions (g/m) | Avg. Weight (Lbs.) | Standard Deviation (g/m) |
|---|---|---|---|
| 2020 | 283.3 | 2,935 | 18.166 |
| 2021 | 277.8 | 2,934 | 14.771 |
| 2022 | 273.2 | 2,960 | 12.795 |
| 2023 | 273.8 | 3,123 | 14.711 |
| 2024 | 273.5 | 3,122 | 12.834 |

The standard deviation data for sedan CO2 emissions appears moderate, with a higher value seen in model year 2020, but then decreasing between model years 2021 and 2024. Overall, the deviation and variance suggest moderate predictability in sedan model carbon emissions. This may be due to the heavier weight of SUVs and the higher performance of the engines.

Table 2 displays average CO2 emissions and weight for 4-cylinder SUV model years 2020-2024. On average, data from table 2 suggests that CO2 emissions are at least 80g/mile greater than emissions from sedan model cars. The average carbon emissions in each model year show slight improvement from 2020 to 2023, and then an increase in 2024. The data also shows that the vehicle weight of 4-cylinder SUVs is at least 800 lbs. greater than the sedan model. Vehicle weight data reveals a slight increase over a five-year period, however nothing significant. An increase in the amount of standard deviation can be observed over the five-year period. The data suggest that there is a higher rate of deviation and variation for the SUVs in the 4-cylinder engine size category, increasing each year. The higher rate of deviation appears to coincide with the increases in CO2 emissions and vehicle weight.

**Table 2: CO2 Emission Data of 4-Cylinder SUVs**

| Year | Avg. CO2 Emissions (g/m) | Avg. Weight (Lbs.) | Standard Deviation (g/m) |
|---|---|---|---|
| 2020 | 366.3 | 3,863 | 28.319 |
| 2021 | 361.7 | 3,855 | 34.839 |
| 2022 | 360.8 | 3,864 | 39.843 |
| 2023 | 359.7 | 3,898 | 43.287 |
| 2024 | 368.5 | 3,948 | 44.672 |

From the information provided for the 4-cylinder SUV category, there is an overall and significant increase in measurable CO2 emissions data observed. Here, a pattern emerges where greater vehicle weight results in a higher rate of carbon emissions from these vehicles. This can be used as an indicator of what the data from the 6-cylinder SUV category will reveal.

Table 3 displays average CO2 emissions and weight for 6-cylinder SUV model years 2020-2024. CO2 emissions in this category appear to remain stable from year to year, with no significant increases or decreases. However, the emissions rate is on average 170g/m more than that of the 4-cylinder sedan models. This is almost exclusively due to the differences in vehicle weight and engine size. An increase in weight of roughly 100g/m is observed for the first three model years, 2020 to 2022, and then stabilizing in 2023 to 2024.

Standard deviation is higher when compared to standard sedan model vehicles, and lower when compared to the 4-cylinder SUVs. When observed through the data, standard deviation increases from 2020 to 2022 and then decreases between 2023 and 2024. This suggests that variation for the 6-cylinder SUV category is high and can be unpredictable.

**Table 3: CO2 Emission Data of 6-Cylinder SUVs**

| Year | Avg. CO2 Emissions (g/m) | Avg. Weight (Lbs.) | Standard Deviation (g/m) |
|---|---|---|---|
| 2020 | 446.5 | 4,703 | 25.575 |
| 2021 | 449.4 | 4,814 | 28.722 |
| 2022 | 456.4 | 4,920 | 40.686 |
| 2023 | 451.2 | 4,958 | 32.272 |
| 2024 | 447.8 | 4,958 | 29.548 |

The increase in observable CO2 emissions data for 6-cylinder SUVs was predictable, despite the standard deviation suggesting otherwise. The relationship between vehicle weight and CO2 emissions is clearly shown in the data displayed in the tables above.

## Hypothesis Testing

The hypothesis of this paper states that SUVs will emit higher levels of CO2. The averages taken from all sample sets of SUV emissions from 2019 to 2024 both 4-cylinder and 6-cylinder will be used for sample mean. The mean CO2 emissions rate for new vehicles (337g/m) will be used to test the hypothesis.

The Null and Alternative Hypotheses are:

**$H_0$: μ≤ 337g/m**

**$H_A$: μ> 337g/m**

A one-tailed hypothesis test will be conducted using α= 0.05; sample size n=10; tα=1.8331

Mean CO2 output for 2020-2024 4-cylinder SUV=363.4g/m and standard deviation= 37.2g/m

If t is greater than 1.8331, then Null Hypothesis is rejected

$t = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}}$ = 26.4g/m÷11.76= 2.24g/m; since t is greater than tα, null hypothesis is rejected.

Mean CO2 output for 2020-2024 6-cylinder SUV=450.3g/m and standard deviation= 30.6g/m

$t = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}}$ = 113.3g/m÷9.68= 11.7g/m; since t is greater than tα, null hypothesis is rejected.

Hypothesis testing for both 4-cylinder and 6-cylinder SUV samples from 2024 supports the alternative hypothesis that SUVs will produce more carbon emissions than standard sedan model passenger vehicles.

## ANOVA Testing

An ANOVA test was conducted to analyze the sample mean of each category of CO2 emissions. Table 4 displays averages of CO2 emissions by model year and engine type and provides an overall average CO2 emissions for the three vehicle categories. The ANOVA test results suggest that all three means of CO2 emissions are not equal. The results reinforce the differences in CO2 emissions between sedan model cars and SUVs.

| **Table 4: Averages of CO2 Emissions by Model Year and Engine Type** | | | |
|---|---|---|---|
| **Model Year** | **Sedan 4-Cyl (g/mile)** | **SUV 4-Cyl (g/mile)** | **SUV 6-Cyl (g/mile)** |
| 2020 | 283.3 | 366.3 | 446.5 |
| 2021 | 277.8 | 361.7 | 449.4 |
| 2022 | 273.2 | 360.8 | 456.4 |
| 2023 | 273.8 | 359.7 | 451.2 |
| 2024 | 273.5 | 368.5 | 447.8 |
| Average | 276.3 | 363.4 | 450.3 |

$\mathbf{H_0: \mu_1 = \mu_2 = \mu_3}$

**$H_A$: Not all means are equal**

Given:

$F_\alpha = 0.05$; k=3; $n_T = 15$; $\bar{\bar{x}} = 363.3\ g/m$; $\bar{x}_1$=276.3g/m; $\bar{x}_2$= 363.4 g/m; $\bar{x}_3$= 450.3 g/m

Formulas Applied:

SST= $\sum_{i=1}^{k} \sum_{j=1}^{n} \left(x_{ij} - \bar{\bar{x}}\right)^2$=75,830; SSB=$\sum_{j=1}^{b} k\left(x_j - \bar{\bar{x}}\right)^2 = 45{,}414$

SSW= SST-SSB= 30,416

$MSW = \frac{ssw}{n_T - k}$= 2,534.67 $MSB = \frac{SSB}{k-1}$=22,707 $F_0 = \frac{MSB}{MSw} = 12$

$F_0$ = 12 > $F_\alpha$ 0.05, 2, 12=3.885, Reject $H_0$ and conclude all means are not equal

Critical Range = $q_{1-\alpha}\sqrt{\frac{MSW}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$ = 84.9g/m

$\bar{x}_1 - \bar{x}_2$=87; $\bar{x}_1 - \bar{x}_3$=174; $\bar{x}_2 - \bar{x}_3$=86.9

When compared to the critical range (84.9 g/m), the differences between all three means are higher. This would verify that all three mean values are not equal.

## Linear Regression Test

A linear regression test was conducted to confirm the relationship between vehicle weight and the rate of CO2 emissions. Table 5 displays the average CO2 emissions and corresponding vehicle weight from each model year and each category (sedan, 4-cylinder SUV and 6-cylinder SUV). This information was used to determine other critical data required to calculate correlation coefficient and is also displayed in Table 5.

**Table 5: Vehicle Weight vs. Emission Rates- Linear Regression Data**

| Model Year | Avg. CO2 Emissions (g/m) | Avg. Weight (Lbs.) | $(x-\bar{x})(y-\bar{y})$ | $(x-\bar{x})^2$ | $(y-\bar{y})^2$ |
|---|---|---|---|---|---|
| 2020 (Sedan) | 283.3 | 2,935 | 79119.70 | 6404.27 | 977461.78 |
| 2021(Sedan) | 277.8 | 2,934 | 84642.89 | 7314.81 | 979440.11 |
| 2022 (Sedan) | 273.2 | 2,960 | 86852.06 | 8122.82 | 928653.44 |
| 2023 (Sedan) | 273.8 | 3,123 | 71681.02 | 8015.02 | 641067.11 |
| 2024 (Sedan) | 273.5 | 3,122 | 72011.04 | 8068.83 | 642669.44 |
| 2020 (4-Cyl. SUV) | 366.3 | 3,863 | -180.38 | 8.84 | 3680.44 |
| 2021 (4-Cyl. SUV) | 361.7 | 3,855 | 111.70 | 2.65 | 4715.11 |
| 2022 (4-Cyl. SUV) | 360.8 | 3,864 | 150.76 | 6.38 | 3560.11 |
| 2023 (4-Cyl. SUV) | 359.7 | 3,898 | 93.08 | 13.15 | 658.78 |
| 2024 (4-Cyl. SUV) | 368.5 | 3,948 | 125.88 | 26.76 | 592.11 |
| 2020 (6-Cyl. SUV) | 446.5 | 4,703 | 64819.75 | 6917.80 | 607360.44 |
| 2021 (6-Cyl. SUV) | 449.4 | 4,814 | 76633.96 | 7408.62 | 792693.44 |
| 2022 (6-Cyl. SUV) | 456.4 | 4,920 | 92732.06 | 8662.65 | 992680.11 |
| 2023 (6-Cyl. SUV) | 451.2 | 4,958 | 90890.32 | 7721.72 | 1069845.44 |
| 2024 (6-Cyl. SUV) | 447.8 | 4,958 | 87373.58 | 7135.74 | 1069845.44 |
| | $\bar{x}$ =363.3 | $\bar{y}$ =3,924 | SUM=807,057.44 | SUM=75830.07 | SUM=8714923.33 |

$\boldsymbol{H_0: \rho = 0}$ $\boldsymbol{(No\ Correlation)}$

$\boldsymbol{H_A: \rho \neq 0}$ $\boldsymbol{(Correlation)}$

Given: n=15, α=0.05, two-tailed test, degree of freedom= 15-2=13, t-critical= 2.1604 (from 't' table)

Formulas: $r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{[\sum(x-\bar{x})^2][\sum(y-\bar{y})^2]}} = 0.9928$ $\quad t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}} = 29.88$

Using the formula for correlation coefficient: , r=0.9928. This result suggests that there is a strong positive correlation between vehicle weight and carbon dioxide vehicle emissions. To rule out sampling error a hypothesis test was conducted, resulting in a 't' value of 29.88. This t-value is greater than the t-critical value of 2.1604, thus the null hypothesis is rejected, and the correlation between vehicle weight and emission rate is validated. The strong positive correlation is illustrated in figure 1 (below).

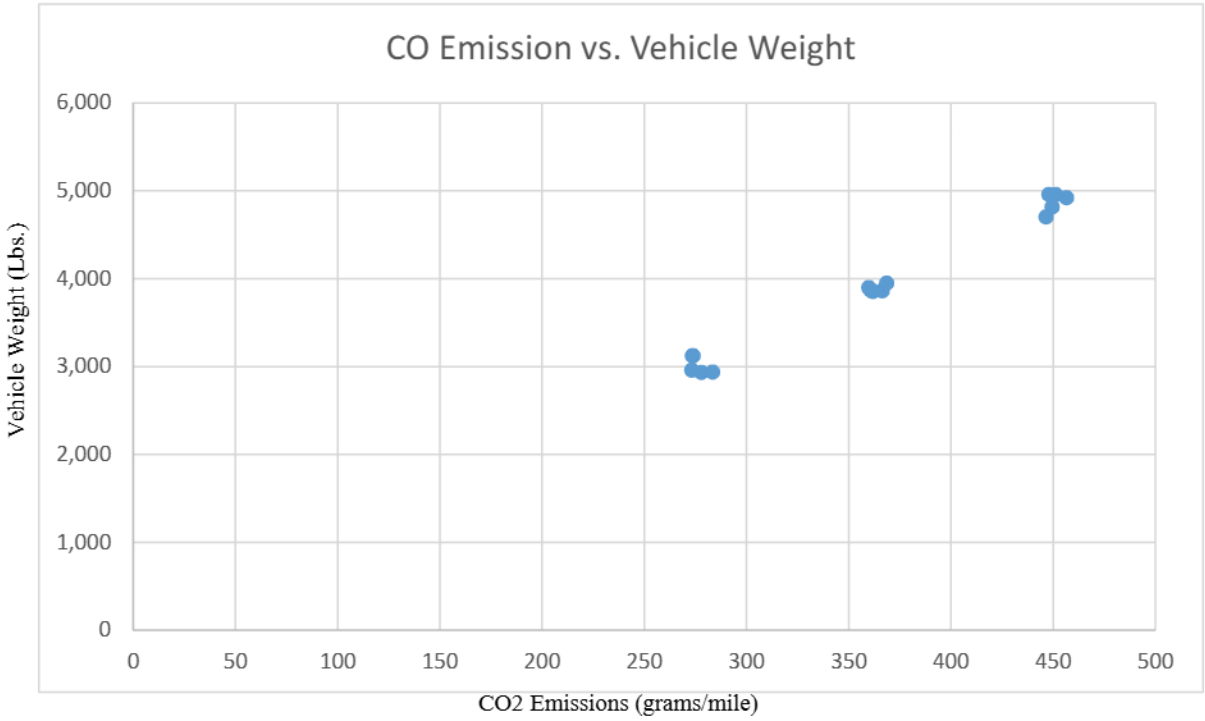



Figure 1: Vehicle Weight vs. CO2 Emission

## Discussion

Through the analysis of the data presented in this paper, it is clear that the shift towards sports utility vehicles has the potential to further add to total emissions in the U.S alone. According to research conducted in the paper "The Rise of SUVs in the US and its Impact on Greenhouse Gas Emissions" (Kovach 2020) additional SUVs and trucks on the road will increase carbon emissions such that projected gains in lowering emissions will be offset by 19-75% [1]. The EPA's 2023 Automotive Trends Report would appear to verify these assertions. According to the report, 63% of new vehicles were trucks or SUVs, and states that "increased weight or horsepower is likely to result in higher CO2 emissions" [3]. This report also mentioned that the shift towards trucks and SUVs in the automotive industry would more than likely offset any gains made in lowering CO2 emissions. To further highlight this trend, General Motors invested over $2 billion in its next generation of trucks and SUVs last year [5]. This investment will ensure that larger vehicles that emit a higher rate of CO2 will roll off the assembly lines well into the next decade.

A recap of the data collected in this paper is as follows. The mean amount of CO2 emissions from 4-cylinder SUVs was found to be approximately 80g/mile greater than that of 4-cylinder sedan type cars. The 6-cylinder SUVs when compared to the standard model car, emits roughly 170g/mile more CO2 gas. Vehicle weight had a proportional relationship to the CO2 emissions in each vehicle category. The mean vehicle weight of 4-cylinder SUVs when compared to standard cars was approximately 800 lbs. greater. And the vehicle weight of the 6-cylinder SUV category was approximately 1,800 lbs. greater when compared to standard sedan cars. The data reflects the fact that heavier vehicles such as the SUV will produce a greater amount of CO2 emissions. It also supports the data from other research done on this topic, as well as claims

made regarding GHG emissions in the referenced research. A study conducted in the United Kingdom ("Sports Utility Vehicles: A Public Health Model of Their Climate and Air Pollution Impacts in the United Kingdom") proved that there exists a positive correlation and linear relationship between vehicle mass and CO2 emissions. Linear regression analysis was conducted in the study and suggested that "vehicle mass was a strong predictor of CO2 emissions" [6]. Data analysis results from this paper regarding vehicle weight and CO2 emissions reflect the findings of the forementioned research conducted in the United Kingdom. A linear regression test conducted in this paper also verified the correlation between vehicle weight and CO2 emissions.

Hypothesis testing verified that CO2 emissions from SUVs will exceed emissions from sedan model cars. An average CO2 emission rate of 337 g/mile was used as a baseline, with the hypothesis test (critical value $\alpha$=0.05) revealing a t-score (2.24) greater than t$\alpha$ (1.8331). This resulted in the rejection of the null hypothesis that CO2 emissions will be less than or equal to baseline emissions (337 g/mile).

The ANOVA test analyzed the overall CO2 emission averages of sedan cars and SUVs (both 4- and 6-cylinder engines); each covering all five model years. A null hypothesis stated that all three categories of CO2 emission rates are equal. The F-test statistic value (12) was found to be larger than the F-critical (3.885), thus resulting in the rejection of the null hypothesis. To further verify the result a Tukey-Kramer multiple comparison was conducted. A critical range of 84.9 was calculated. When compared to the differences between sample means from each population (category) it was found that each resulting value was larger than the critical range. This is verification that all three averages are not equal. The largest gap when compared to critical range is the difference between sedan cars and 6-cylinder SUVs (174 vs 84.9). This also highlights the significant rate of 6- cylinder SUV CO2 emissions when compared to sedan model cars.

## Conclusion

Fuel economy in the United States has seen significant improvement over the past four decades. However, the automotive industry's shift away from standard sedan model cars and towards sports utility vehicles, has been counterproductive in this effort. The number of sedan cars on the road has halved over the years, from 80 to 41%. The data presented in this paper validated the hypothesis that heavier vehicles such as SUVs will produce greater amounts of carbon dioxide gas than standard passenger cars. ANOVA testing further explored and validated the significant difference in carbon dioxide emissions compared to standard sedan model cars. The vehicle weight of SUVs is on average 1.5 to 2 times greater than that of standard cars.The correlation between vehicle weight and rate of emissions was verified through a linear regression test. The rate of CO2 emissions from SUVs was found to be up to 1.6 times that of standard sedan cars. The data also reflects and supports the information presented in the research material cited in this study. Research identifies emissions from automobiles as the leading contributor to greenhouse gas (GHG) emissions in the U.S., with SUVs producing the majority of these gases. The automotive industry continues to invest in larger, heavier vehicles that emit higher amounts of GHGs. Unless effective mitigation measures are taken, levels of GHGs including CO2 will only continue to rise and negatively impact the environment and human health.

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